CSCE 4430 – Programming Languages

Active Learning 3 – Names, Types, Type Systems, Semantics

Micheal Landsbaum

11408297

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1.

A. In C-like languages, identifiers are case sensitive (e.g., alpha and Alpha can name different entities)

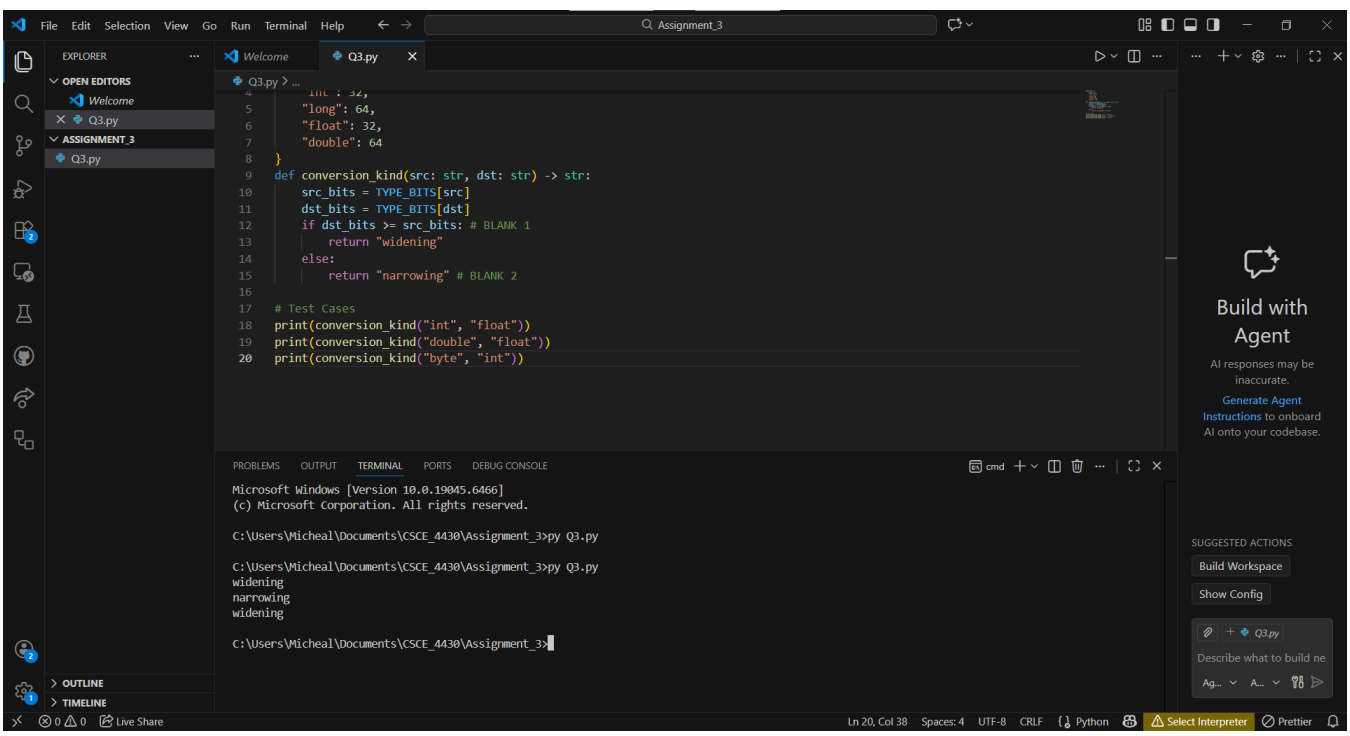
B. Reserved words / keywords carry special meaning and cannot be used as identifiers.

D. A language is strongly typed if type errors can be detected either at compile time or at run time.

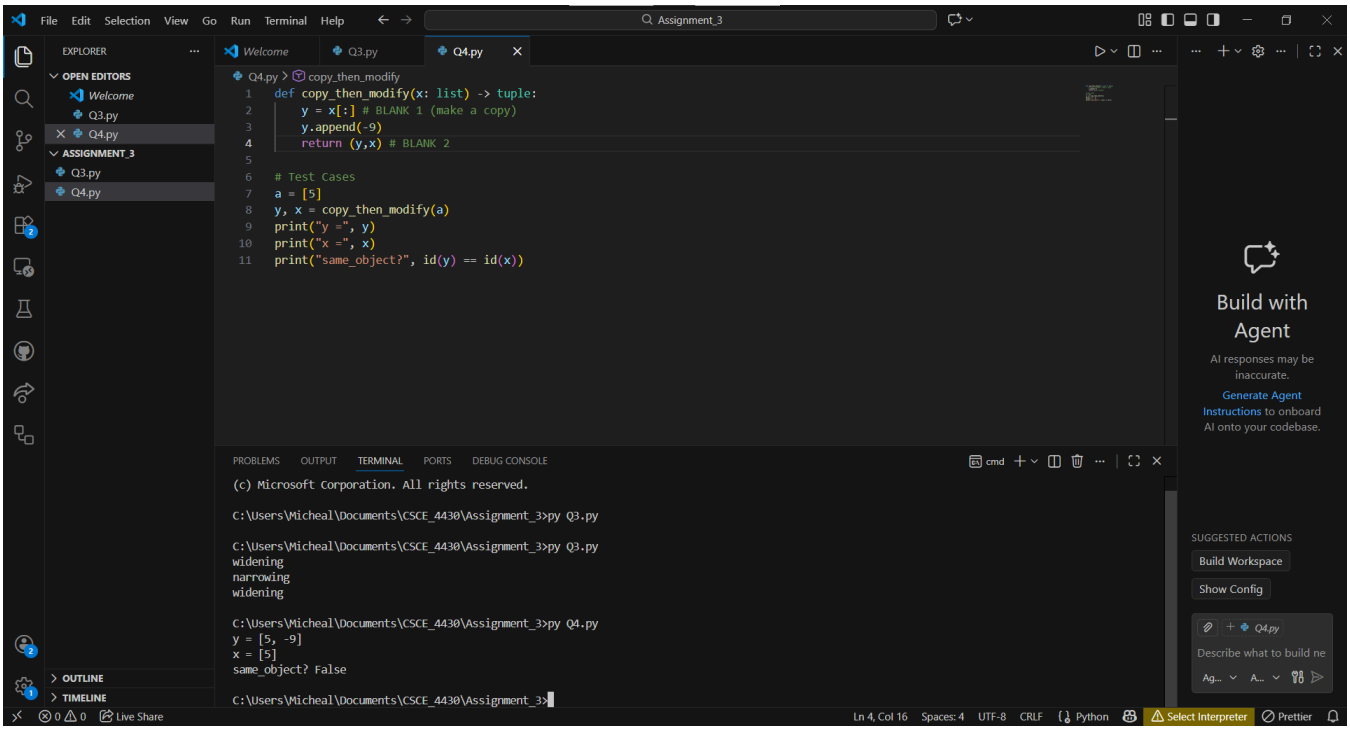
2.

- Value semantics changes actual data stored
- Reference semantics change the address or pointer

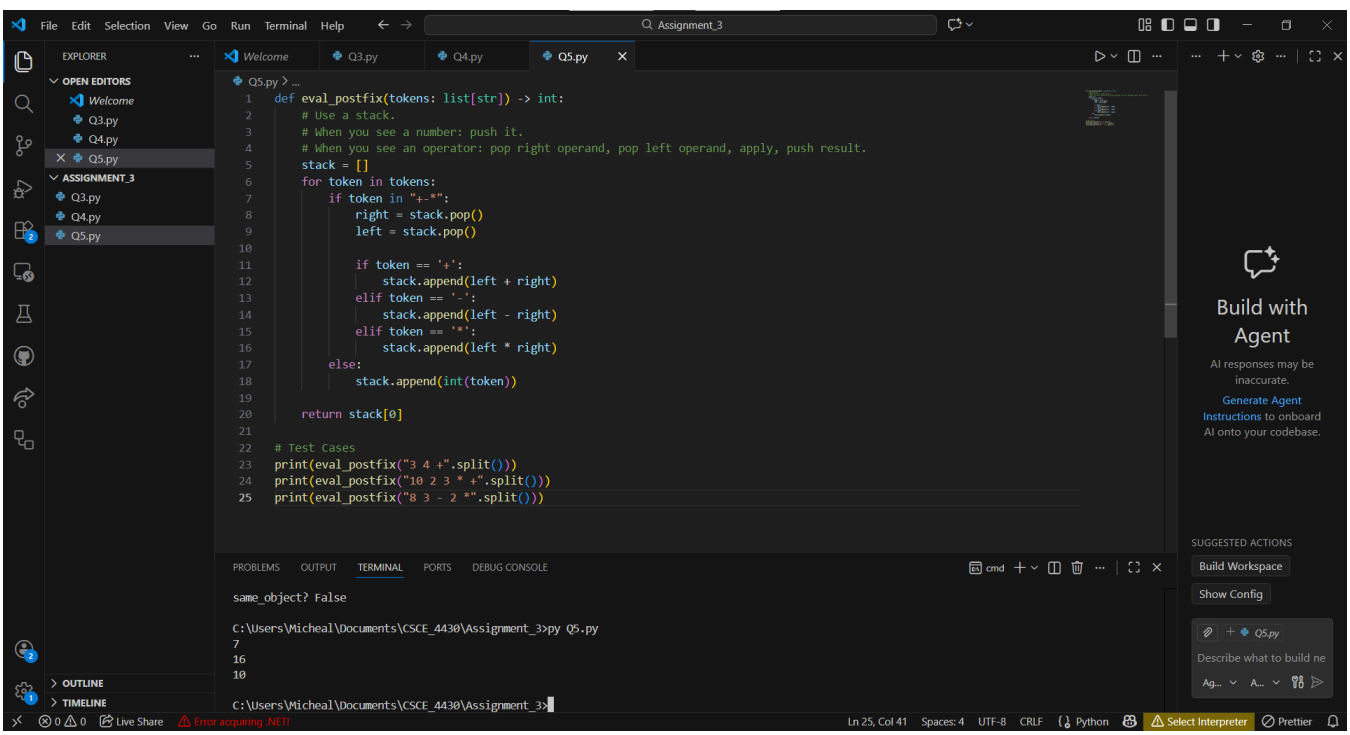
3.



4.


File Edit Selection View Go Run Terminal Help
Assignment_3
EXPLORER
OPEN EDITORS
Welcome
Q3.py
Q4.py
ASSIGNMENT_3
Q3.py
Q4.py
Q4.py > copy_then_modify
def copy_then_modify(x: list) -> tuple:
y = x[:] # BLANK 1 (make a copy)
y.append(-9)
return (y,x) # BLANK 2
# Test Cases
a = [5]
y, x = copy_then_modify(a)
print("y =", y)
print("x =", x)
print("same_object?", id(y) == id(x))
PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
(c) Microsoft Corporation. All rights reserved.
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>py Q3.py
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>py Q3.py
widening
narrowing
widening
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>py Q4.py
y = [5, -9]
x = [5]
same_object? False
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>
Build with Agent
AI responses may be inaccurate.
Generate Agent Instructions to onboard AI onto your codebase.
SUGGESTED ACTIONS
Build Workspace
Show Config
Describe what to build ne
Ln 4, Col 16 Spaces: 4 UTF-8 CRLF Python Select Interpreter Prettier

5.


File Edit Selection View Go Run Terminal Help
Assignment_3
EXPLORER
OPEN EDITORS
Welcome
Q3.py
Q4.py
Q5.py
ASSIGNMENT_3
Q3.py
Q4.py
Q5.py
Q5.py > ...
def eval_postfix(tokens: list[str]) -> int:
# Use a stack.
# When you see a number: push it.
# When you see an operator: pop right operand, pop left operand, apply, push result.
stack = []
for token in tokens:
if token in "+-*":
right = stack.pop()
left = stack.pop()
if token == '+':
stack.append(left + right)
elif token == '-':
stack.append(left - right)
elif token == '*':
stack.append(left * right)
else:
stack.append(int(token))
return stack[0]
# Test Cases
print(eval_postfix("3 4 +".split()))
print(eval_postfix("10 2 3 * +".split()))
print(eval_postfix("8 3 - 2 *".split()))
PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
same_object? False
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>py Q5.py
7
16
10
C:\Users\Micheal\Documents\CSCE_4430\Assignment_3>
Build with Agent
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Generate Agent Instructions to onboard AI onto your codebase.
SUGGESTED ACTIONS
Build Workspace
Show Config
Describe what to build ne
Ln 25, Col 41 Spaces: 4 UTF-8 CRLF Python Select Interpreter Prettier